

Quiz 3

Sep 13th, 2024

Time: 15 minutes

Name _____

Roll no _____

Group: _____

Instructions: Notes, books, computers, cell phones and other electronic devices are not allowed.
Max marks = 5.

Problem 1. 1. Find an LU factorization of the following matrix.

$$A = \begin{bmatrix} 7 & -1 & 0 \\ 14 & 0 & 1 \\ 7 & -3 & 3 \end{bmatrix}$$

2. Using the above factorization, solve the linear system $A\bar{x} = \bar{b}$ where $\bar{b} = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$

Note: You are not allowed to use any other method to solve the system.